

2.2 Request Path (Development)

Component	Port	Role
Vite dev server	3000	UI, proxies `/api` ! 3001
Express `dev:api`	3001	JSON API, reads SQLite + Postgres
`pnpm start` (prod)	3000	Bundled static + API (or API-only if no...

2.3 Shared Logic

Business rules live in shared/ so the same comp-matching, flip math, TCAD address scoring, and DTO composition run in scripts, API routes, and (where applicable) the client via @shared alias.

3. Repository Layout

Path	Purpose
`client/`	React UI (Comparables landing, flip results, shadcn...
`server/`	Express app, route handlers
`shared/`	Domain logic: MLS, TCAD, comparables, property-profile, flip
`scripts/`	Seed jobs (`seed-mls`, `seed-mls-open`, `seed-tcad`), `smoke...
`data/`	Gitignored runtime data (`mls.sqlite`, raw JSON dumps)
`docs/`	Roadmap and architecture documentation

4. Data Sources & Storage

4.1 MLS (Multiple Listing Service)

Aspect	Detail
Upstream	MLS Grid API (`https://api-demo.mlsgrid.com/v2` or production...)
Originating system	`actris` (Austin Board of Realtors) — `shared/mls/constants.ts`
Local store	SQLite `data/mls.sqlite`, table `properties`
Seed commands	`pnpm seed:mls` (closed residential), `pnpm seed:mls:open`...
Runtime	API reads SQLite only; no live MLS Grid on user requests
Token	`MLS_GRID_TOKEN` or `VITE_MLS_GRID_TOKEN` in...

Row shape: normalized address, lat/lon, beds/baths/sqft, list/close price, close date, pool, garage, lot acres, property condition, status.

4.2 TCAD (Travis Central Appraisal District)

Aspect	Detail
Upstream	ArcGIS MapServer layer `TCAD/MapServer/0` (full county...)
Local store	Postgres `tcad_parcel` (Neon) via `DATABASE_URL`
Seed command	`pnpm seed:tcad` (~387k parcels, paginated ETL)
Runtime	Cache-first in `shared/tcad/client.ts`; live ArcGIS fallback
Coverage	Travis County only — Williamson/Georgetown addresses retur...

Fields cached: PROP_ID, geo_id, situs address, tax values (appraised, market, assessed), acres, WGS-84 centroid.

4.3 Data Independence Principle

MLS and TCAD are intentionally separate. There is no required listing_key !" prop_id crosswalk in v1. Profiles compose both sources at request time when possible; partial profiles are valid.

5. Core User Flows

5.1 Comparables Search

1. User enters address (with typeahead from MLS SQLite). 2. UI calls GET /api/comparables/unified with radius, recency, optional includeTcad. 3. Server resolves subject from MLS and/or TCAD. 4. Returns three optional sections: - Closed sales (MLS) — primary sale comps - Active listings (MLS open status) - Tax references (TCAD radius) — reference only, not sale comps

Comps are bucketed into exclusive mile rings (0–0.5, 0.5–1, 1–2 mi, ...) with match % from beds/baths/distance.

5.2 Property Profile

1. GET /api/property/profile?address= or ?propId= 2. fetchPropertyProfile() loads MLS by address and TCAD by address or propId. 3. composePropertyProfile() merges identifiers, physical, MLS sale, tax, location. 4. TCAD is always attempted for tax data even when MLS supplies coordinates. 5. Ambiguous TCAD matches surface as taxCandidates rather than silent failure.

5.3 Flip Analysis

1. User runs comparables search, switches to Flip analysis tab. 2. Enters purchase price and rehab scope tier (cosmetic / moderate / full). 3. POST /api/predict/flip in enriched mode: - Loads property profile (MLS + TCAD) - Derives ARV from similar closed MLS comps (median \$/sqft × subject sqft) - Computes rehab, closing, hold (tax basis from assessed/appraised), financing - Returns viability band: strong / marginal / weak / negative
Manual mode: user supplies explicit arv + livingAreaSqft without subject lookup.

6. API Surface

Method	Endpoint	Purpose
GET	/api/health`	Liveness check
GET	/api/properties/suggest`	Address typeahead (MLS)
GET	/api/properties/by-address`	Single MLS property
GET	/api/properties/by-radius`	Closed comps in radius
GET	/api/comparables/unified`	Closed + open + optional TCAD
GET	/api/property/profile`	Composed MLS + TCAD profile
GET	/api/tcad/property`	TCAD parcel by propId or address
POST	/api/predict/flip`	Rules-based flip prediction

All routes are mounted in server/app.ts.

7. Domain Modules

7.1 shared/mls/

- RESO property types, MLS Grid client, address normalization - scripts/mls-db.ts: SQLite schema, radius queries, address resolution

7.2 shared/tcad/

- ArcGIS client with retry/timeout - Address fuzzy matching (address-query.ts) - Postgres cache (db.ts) - Parcel centroids from polygon geometry

7.3 shared/comparables/

- match-score.ts — mile rings, match %, distance score - comp-record.ts — unified CompRecordDto with source and compRole - unified-search.ts — orchestrates MLS + TCAD comp sections

7.4 shared/property-profile/

- fetch-profile.ts — enrichment orchestration - compose.ts — merges partial MLS/TCAD into PropertyProfileDto - Provenance and completeness scoring

7.5 shared/flip/

- config.ts — Travis County rehab tiers, closing %, hold/financing defaults - arv-comps.ts — similarity filter, tier-based \$/sqft slice (top half / quartile) - deal-costs.ts — deterministic cost stack - predict-flip.ts — end-to-end prediction pipeline

8. Flip Prediction Engine

8.1 ARV (After Repair Value)

Source: MLS closed sales within radius, filtered by:

- Recency (maxAgeMonths on close_date) - Similarity: $\pm 25\%$ sqft, beds/baths within delta rules - Rehab tier slice: cosmetic uses all similar; moderate uses top 50% \$/sqft; full uses top 25%

ARV = median(pricePerSqft of slice) $\times$ subjectLivingAreaSqft

Override: request body arv !' manual mode.

8.2 Cost Stack

Component	Basis
Rehab	tier \$/sqft $\times$ living area
Buy closing	% of purchase price
Hold	property tax (assessed !' appraised !' purchase), insurance,...
Financing	hard-money style: LTC ratio, points, interest during hold
Sell closing	% of ARV

8.3 Viability Bands

Net margin % (of ARV)	Label
"e 12%	strong
6–12%	marginal
0–6%	weak
< 0%	negative

8.4 Explicit Non-Goals

- TCAD appraised value is not used as ARV or sale comp - Tax values affect hold cost only (when TCAD resolves)

9. Key Design Decisions

Decision	Rationale
MLS in SQLite, not live API	Fast radius queries; avoids MLS Grid rate limits on every request
TCAD in Postgres (Neon)	~387k parcels; eliminates ArcGIS timeouts on hot paths
Cache-first TCAD with ArcGIS fallback	Works before seed completes; resilient to cache gaps
No forced MLS!TCAD link in v1	Address matching is messy; partial profiles OK; crosswalk is...
TCAD tax comps are reference only	Appraised "" market sale price; prevents misleading ARV
Rules engine before ML	Auditable math; same API contract for future Python model
Shared TypeScript domain layer	One source of truth for comps, flip, and profile logic
Enriched vs manual flip modes	Supports full subject lookup or spreadsheet-style what-if
Dedupe PROP_ID on TCAD seed	ArcGIS pages can repeat prop_ids (multipart parcels)

10. Operational Commands

```
pnpm install
pnpm seed:mls          # MLS closed !' data/mls.sqlite
pnpm seed:mls:open     # Active/pending into same DB
pnpm seed:tcad         # Full county !' Neon tcad_parcel
pnpm dev:api           # API :3001
pnpm dev               # UI :3000
pnpm smoke:flip        # API/engine regression checks
pnpm build && pnpm start # Production-style single server
```

Environment (.env.local):

- VITE_MLS_GRID_TOKEN / MLS_GRID_TOKEN — MLS Grid bearer token - DATABASE_URL — Neon Postgres for TCAD cache

11. Roadmap & Gaps

Completed (Phases 1–4 + partial 5)

- MLS schema extensions, mile-bucket comps, unified comparables API - Property profile composition, TCAD propId/address APIs - Flip rules engine v0, flip UI on comparables landing - TCAD Postgres ETL (seed:tcad), cache-first reads - TCAD skip fix (always fetch tax when enriching by address)

Planned

Item	Description
MLS !' TCAD crosswalk	Precomputed `listing_key` !' `prop_id` links
Deal ledger	Labeled flips for ML training
Python ML ARV model	Same `POST /api/predict/flip` contract
`geold` lookup	Alternate TCAD key
Monthly TCAD refresh	Cron re-run `seed:tcad`

Known Limitations

- MLS seed cap (~2k–5k rows configurable) — not full ACTRIS history - Non-Travis addresses have MLS data but no TCAD tax - Flip ARV requires sufficient similar closed comps in radius - Home page calculator may still use embedded constants if MLS token unset

12. Appendix: Diagrams

12.1 Property Profile Composition

12.2 Flip Prediction Pipeline

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12.3 TCAD Read Path

12.4 Comp Roles (Unified API)

[illegible]

End of report. Source: Realty360-Austin codebase and docs/ROADMAP.md.